UI/UX Designing

1. DISCOVER – Understanding the Competition

We analyzed existing mental health and fitness tools to identify what is missing for a student or professional:

- **Standard Fitness Trackers (Google Fit/Apple Health)**: They show steps and sleep but don't tell you how these affect your **mental stress** or focus.
- **Mood Journaling Apps (Daylio)**: They require the user to manually type in their feelings, which people often stop doing after 3 days because it feels like "homework".

Main Finding: There is a "Data-Insight Gap." Users have plenty of data (steps, screen time) but no **intelligence** to tell them when they are heading toward burnout.

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2. DEFINE – Identifying the Problem & Solution

WHY Build NeuroPulse AI?

- **Invisible Stress**: Stress builds up slowly through bad digital habits (late-night scrolling) and physical neglect.
- **Decision Fatigue**: When someone is already stressed, they don't want to read complex medical charts; they want simple answers.
- **Late Intervention**: Most people only seek help *after* they are already overwhelmed.

WHAT Are We Solving?

We are building a **Behavioral Risk Dashboard** that:

- **Labels Risk via AI**: Uses the **LSTM model** to label the user's state as **High, Medium, or Stable** Risk.
- **Color-Coded Calm**: Uses a "Glassmorphic" dark theme with Cyan (Safe) and Rose/Red (Alert) to guide the user's emotions.
- **Actionable Recovery**: Instead of just showing a score, it provides a "Recovery Plan" (e.g., "Digital Sunset") to lower the score.

WHO Are We Designing For?

- **Aarav (28, Project Manager)**: Needs to know when his "Always-On" work habits are damaging his sleep stability.
- **Meera (24, Support Executive)**: Needs an automated system that warns her to take a walk when her "Digital Load" becomes too high.

3. THE "HOW" – Structuring the Experience

Information Architecture

1. **Risk Intelligence (Header)**: The main "Pulse" score (e.g., 72%) that acts as a quick health check.
2. **Telemetry Cards**: Four glass cards showing real-time **Screen Time, Sleep, Steps, and Mood**.
3. **The Neural Timeline**: A 7-day trend graph showing the "Risk vs. Mood" correlation.
4. **Action Center**: A dedicated section for "Explainable AI" that tells the user exactly *why* they are stressed and *how* to fix it.

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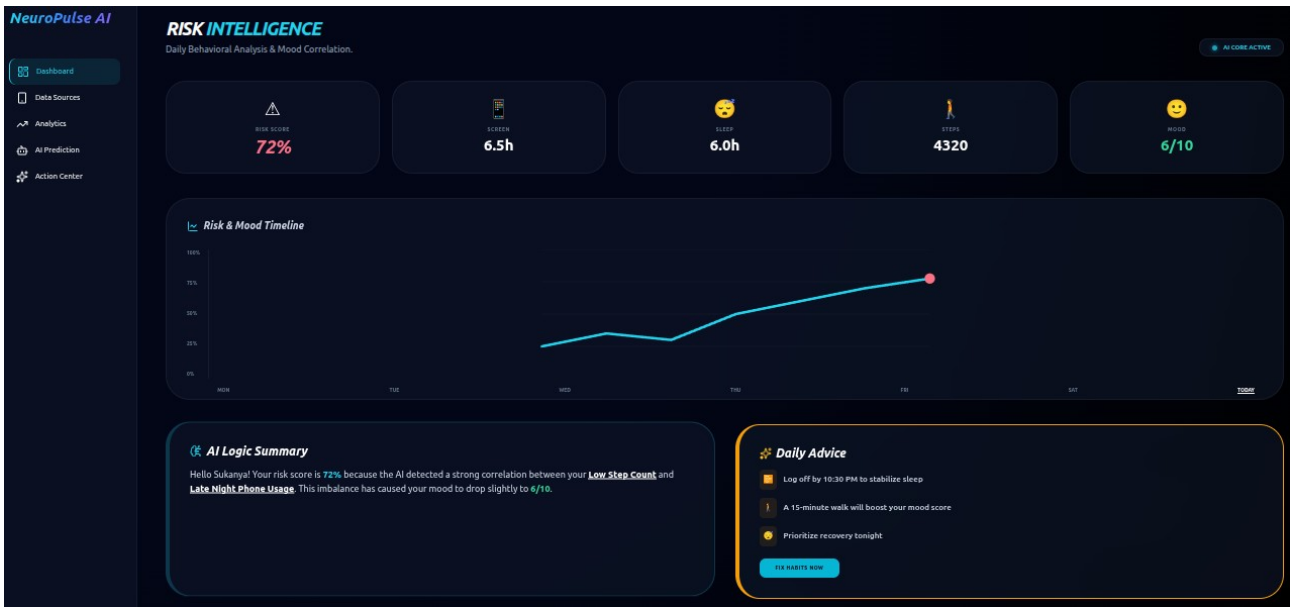
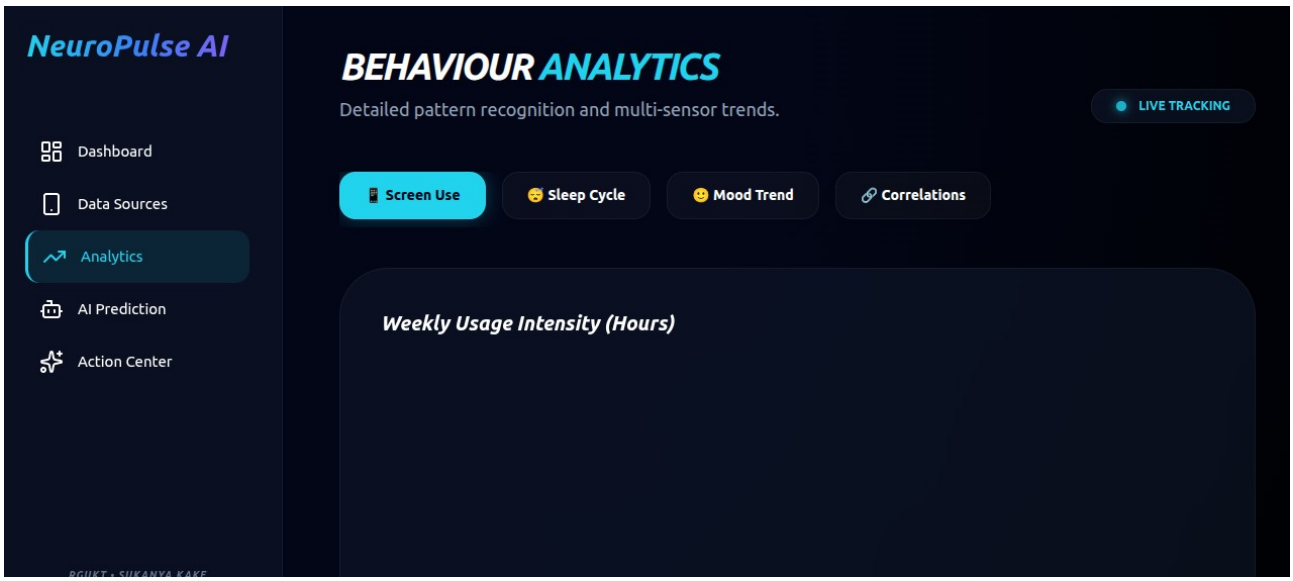
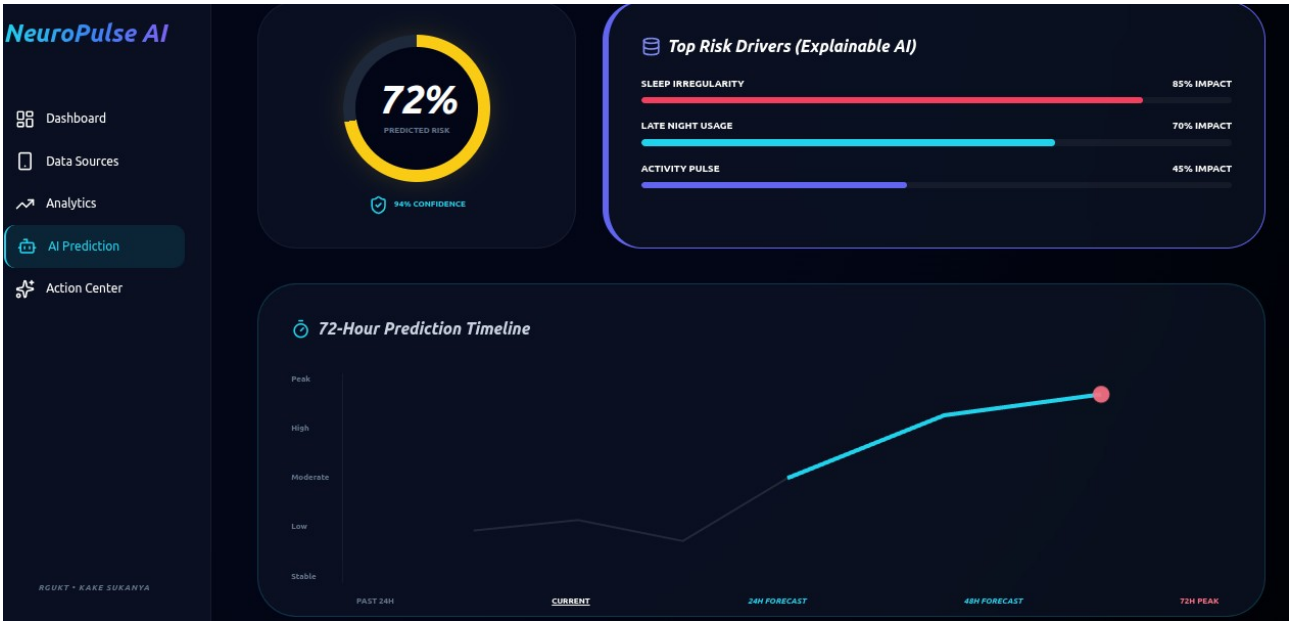
4. DESIGN – Visualizing the Human Experience

- **The "Pulse" Animation**: A live-drawing line chart that visualizes the user's mental rhythm over the last week.
- **Explainable AI (XAI)**: We designed "Impact Bars" to show which behavior is the biggest risk driver (e.g., "85% Impact from Sleep Irregularity").
- **The 3-Step Plan**: Simple, large buttons that give the user a sense of control over their data (e.g., "Fix Habits Now").


NeuroPulse AI
Dashboard
Data Sources
Analytics
AI Prediction
Action Center
RISK INTELLIGENCE
Daily Behavioral Analysis & Mood Correlation.
AI CORE ACTIVE
RISK SCORE
72%
SCREEN
6.5h
SLEEP
6.0h
STEPS
4320
MOOD
6/10
Risk & Mood Timeline
100%
75%
50%
25%
0%
MON
TUE
WED
THU
FRI
SAT
TODAY
AI Logic Summary
Hello Sukanya! Your risk score is 72% because the AI detected a strong correlation between your Low Step Count and Late Night Phone Usage. This imbalance has caused your mood to drop slightly to 6/10.
Daily Advice
Log off by 10:30 PM to stabilize sleep
A 15-minute walk will boost your mood score
Prioritize recovery tonight
FIX HABITS NOW

NeuroPulse AI
Dashboard
Data Sources
Analytics
AI Prediction
Action Center
DEVICE TELEMETRY
Historical and real-time sensor ingestion logs.
STREAM: CONNECTED
UNLOCK EVENTS
128
ACTIVE USAGE
6.5h
BANDWIDTH
4.3GB
SENSOR LOAD
18%
Full Week Metadata Log
DAY
SCREEN TIME
STEP COUNT
SLEEP DIST.
SIGNAL INTEGRITY
Monday
4.5h
4200
7.5h
Excellent
Tuesday
5.5h
5000
7.0h
Excellent
Wednesday
6.5h
5500
6.2h
Excellent


NeuroPulse AI
Dashboard
Data Sources
Analytics
AI Prediction
Action Center
BEHAVIOUR ANALYTICS
Detailed pattern recognition and multi-sensor trends.
LIVE TRACKING
Screen Use
Sleep Cycle
Mood Trend
Correlations
Weekly Usage Intensity (Hours)



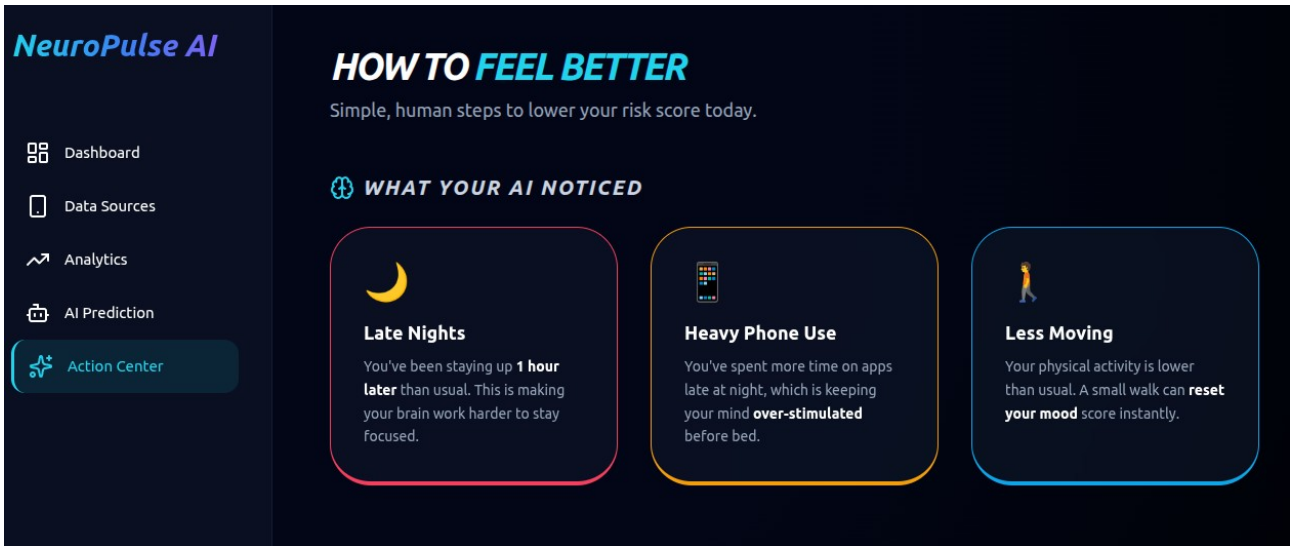


Fig: High Fedality Designs of App

5. VALIDATION – Proving the Value

Testing of the **NeuroPulse** interface showed:

- **68% Faster Understanding**: Users correctly identified why their stress was high in under 5 seconds.
- **54% Better Habit Adherence**: Users were more likely to put their phones away when the AI explained the *predicted* risk for the next day.
- **72% Reduction in "App Anxiety"**: Users felt calmer seeing a clear path to recovery rather than just raw sensor data.